

For my array, linked list, stack, queue, and deque, I created testing suites for each structure. I tested every function that manipulated the data structure in some way, and calculated the correct answer via a whiteboard to ensure I tested for the correct answer, then provided it to the "Assert::AreEqual" function for each test method. Most with more than one AreEqual test. I will focus on the array since it had the most tests. For the dynamic array, I tested for, AddElement, InsertElement, GetSize, GetCapacity, IsEmpty, CopyConstructor, GetData, RemoveData, ShrinkToFit, and ClearData. For "AddElement" I created an array of ints and another array of floats. I added some elements via the "Add" function, and checked each element against the element I hand provided which resulted in a passing grade. For "InsertElement" I inserted an int to an array in the middle of the array, ensuring that the array correctly resized and put the correct element in the spot. Same thing for a float array. For "RemoveData" I created an array of ints, set all elements, then randomly removed an index, and checked every other index against the index removed, to ensure that the array correctly removed the index. I did it once for Preserving the Order and once for not preserving the order. Overall, the tests accurately reflected the reason for each function, and correctly showed flaws in the code for fixing, and ensured that my Dynamic classes are properly setup for production.